

Reaction–diffusion and Turing patterns

How diffusion can destabilize a uniform membrane

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Diffusion smooths each concentration field in isolation. Coupled to reactions, unequal diffusion can instead destabilize a homogeneous state. The essential calculation is a competition between the reaction Jacobian and the spatial Laplacian spectrum. This note starts with two species on a fixed, locally flat membrane.

1 Separate local chemistry from spatial transport

Let $u(\mathbf{x}, t)$ and $v(\mathbf{x}, t)$ be surface concentrations, with reaction rates $f(u, v)$ and $g(u, v)$ and constant positive diffusivities D_u, D_v :

$$\partial_t u = f(u, v) + D_u \nabla^2 u, \quad \partial_t v = g(u, v) + D_v \nabla^2 v. \quad (1)$$

Choose a homogeneous steady state (u_0, v_0) with $f = g = 0$. All derivatives below are evaluated there. Its local reaction Jacobian is

$$J = \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix}, \quad \text{tr } J = f_u + g_v < 0, \quad \det J = f_u g_v - f_v g_u > 0. \quad (2)$$

These inequalities make the well-mixed state linearly stable. They are essential: a spatial pattern developing from an already unstable local reaction system need not be a Turing instability.

A perturbation proportional to $e^{st+i\mathbf{q}\cdot\mathbf{x}}$ evolves with

$$J_q = J - q^2 \begin{pmatrix} D_u & 0 \\ 0 & D_v \end{pmatrix}, \quad q = |\mathbf{q}|. \quad (3)$$

Since $\text{tr } J_q = \text{tr } J - (D_u + D_v)q^2 < 0$, instability requires the determinant to become negative:

$$\det J_q = \det J - (D_v f_u + D_u g_v)q^2 + D_u D_v q^4. \quad (4)$$

Thus the instability is stationary at onset in this two-species, diagonal-diffusion model; it is not a finite-wavenumber Hopf bifurcation.

2 The classical Turing conditions

The quadratic polynomial in q^2 in Eq. (4) has a negative minimum at positive q^2 precisely when

$$D_v f_u + D_u g_v > 0, \quad (D_v f_u + D_u g_v)^2 > 4D_u D_v \det J. \quad (5)$$

Together with Eq. (2), these are the continuum-wavenumber conditions for a diffusion-driven instability. If $D_u = D_v = D$, then $J_q = J - Dq^2 I$: every growth rate shifts left, and the instability is impossible. Cross-diffusion or additional species can change this conclusion.

3 An example and a geometry test

In dimensionless units, take a local linearization and diffusion matrix

$$J = \begin{pmatrix} 1 & -2 \\ 2 & -3 \end{pmatrix}, \quad D_u = 1, \quad D_v = 10. \quad (6)$$

Then $\text{tr } J = -2$, $\det J = 1$, and

$$\det J_q = 1 - 7q^2 + 10q^4 = (1 - 5q^2)(1 - 2q^2). \quad (7)$$

The unstable band is therefore $1/5 < q^2 < 1/2$. For example, $q^2 = 0.3$ gives $\det J_q = -0.2$ and one positive growth rate. These numbers specify a linear example; nonlinear reaction terms are still needed to determine the final pattern and keep concentrations physical.

A finite domain does not permit arbitrary q . On a periodic one-dimensional interval of length L , $q_n = 2\pi n/L$; at least one allowed nonzero mode must enter the unstable band. Local kinetics satisfying Eq. (5) can therefore remain stable in a domain too small to fit an unstable mode.

On a fixed spherical membrane of radius R , replace the planar Laplacian by the Laplace–Beltrami operator. Spherical harmonics satisfy

$$-\nabla_S^2 Y_{\ell m} = \frac{\ell(\ell+1)}{R^2} Y_{\ell m}, \quad \ell = 0, 1, 2, \dots \quad (8)$$

The same determinant test applies with $q^2 = \ell(\ell+1)/R^2$. Radius selects which angular modes can grow. Curvature enters through this mode spectrum in the present model; curvature-dependent chemistry or membrane deformation would add further couplings.

4 The biological model determines the conservation law

Local self-activation and a faster-spreading inhibitory influence provide a common realization of the inequalities. But not every intracellular pattern is a classical two-species activator–inhibitor pattern. A particularly important distinction is whether the total amount of protein is conserved.

If the two species interconvert with $g = -f$, then

$$\partial_t(u+v) = D_u \nabla^2 u + D_v \nabla^2 v, \quad \frac{d}{dt} \int (u+v) dA = 0 \quad (9)$$

under no-flux boundaries. The homogeneous Jacobian has $\det J = 0$: the conserved total produces a neutral uniform mode, so the strict stability conditions in Eq. (2) are no longer the right starting point. Spatial redistribution can still destabilize the system, but must be analyzed at fixed total mass [2].

Actual membrane–cytosol exchange couples surface and volume concentrations and often involves more than two chemical states. A two-dimensional surface model is a reduction whose assumptions should be checked. Linear instability predicts early growth and admissible wavelengths; it does not by itself select spots, stripes, traveling waves, or a unique final state.

References

- [1] A. M. Turing, “The Chemical Basis of Morphogenesis,” *Phil. Trans. R. Soc. B* **237**, 37–72 (1952). doi:10.1098/rstb.1952.0012.
- [2] F. Brauns, J. Halatek, and E. Frey, *Phase-space geometry of mass-conserving reaction–diffusion dynamics*. arXiv:1812.08684.